Cole Palmeri and Noah Reichman

Final Paper Independent Study

Winning and Player Compensation During the Moneyball and Shift Eras and Its Implications for MLB's New Rule Changes

Abstract:

The 2023 season will be a milestone in Major League Baseball's (MLB) history. With the implementation of the shift ban, bigger bases, and the pitch clock at the beginning of the 2023 season, baseball is undergoing a massive change. But, we've seen changes to the game before, especially within the past 20 years. How do these changes affect what matters to winning and player compensation? We will analyze two periods in Major League Baseball: the Moneyball Era (2000-2010) and the Shift Era (2011-2022). In each period, we examined what contributes to winning, and if what contributes to winning is compensated accordingly in players' salaries. After analyzing this, we predicted what will matter more in winning and changes in player compensation with the new rule changes. We discovered that what matters to winning does change depending on the period and most of these changes do show up in player compensation, directly or indirectly.

Introduction:

This paper will analyze two time periods in Major League Baseball: the Moneyball Era and the Shift Era. For the sake of this study, we will broadly define the Moneyball Era as a period from 2000-2010 and the Shift Era from 2011-2023. Our goal for the paper is to identify

what is important to winning for MLB teams and see if players are compensated accordingly. We also want to see if certain statistics are more important to winning and player compensation in different periods. From there, we will predict what will be more important in winning and in players' salaries with MLB's new rule changes.

Other Literature on Related Topics:

Herman Demmink III (2010) analyzed the importance of stolen bases and on-base percentage (OBP) on the number of wins. He examined the 1990-2004 seasons, so there will be some overlap with our analysis time frame. He found that both stolen bases and OBP are important predictors of wins. Demmink also examined the effect of getting caught stealing. He found that the coefficient was -21.7, meaning that for every caught stealing, a team lost -21.7 wins, ceteris paribus. However, Demmink acknowledges some of the limitations of his study. He states that the timing of the steal within the game matters a lot (Demmink, 2010).

Christopher B. Shorey (2016) found a statistically significant relationship between defensive runs saved (DRS) and a player's salary for the 2012-2015 seasons. He found that a one-unit increase in DRS per year leads to a 3.5% increase in a player's salary (Shorey, 2016). This has implications not only for a player's salary but also for what matters to winning. We can expect to see a positive correlation between DRS and winning percentage (WP) in the data, but this finding may vary depending on the period we are analyzing.

Congdon-Hohman and Lanning (2018) did a study that examined how player compensations have changed in the MLB over time. They chose three sets of three-year periods to conduct their study: 1988-1990, 1998-2000, and 2008-2010. By looking at how aggregate statistics like slugging percentage (SLG), on-base percentage (OBP), and on-base plus slugging (OPS), and counting stats like home runs, walks, doubles, and hits adjusted player

compensations over these periods, showed the way players are compensated does change over time as to what teams think is important during that time. This is important because we are trying to find if the Shift Era adjusted the way teams win and if players were compensated accordingly.

Hakes and Sauer (2006) did a study very similar to ours for the years 2000-2004. They wanted to confirm the Moneyball hypothesis. The analysis started with finding what is important to winning, then looked to see if what is important to winning is compensated accordingly in players' salaries. They found that OBP was more important than slugging percentage to a team's winning percentage, but SLG was valued more in salaries. This was until 2004 when there was a shift in player salaries and the market overcorrected with the OBP coefficient increasing from the previous years, and being greater than the SLG coefficient. This means that OBP became valued higher than SLG in player salaries, and the market corrected to compensate players according to what is important to winning.

We plan to use all of the aforementioned studies to not only make predictions about what our models will say but also for developing our model and what variables to include. We decided to include OBP, SLG, and DRS along with other variables that were included in the models discussed above.

Background:

2023 MLB Rule Changes:

The MLB put in three new rule changes for the 2023 season. All of these changes were put into place to bring more action to the sport and to increase the pace of play. The first rule change is bigger bases. A base was traditionally a 15-inch square, now it is an 18-inch square. The primary goal of the base size increase is to limit collisions and give players more room to

operate around the bases. A secondary goal of the larger bases is to increase stolen base attempts and make the game more offensive. The second rule change is the shift ban. The banning of the shift is supposed to make the game more offensive and appealing to watch. With the new rule change, two infielders have to be on either side of second base and all infielders have to stay on the infield, which in theory should allow more hits to happen, making the game more exciting. The third rule change this year is the pitch clock. The pitch clock was implemented to speed up the pace of play. Pitchers now have 15 seconds to deliver a pitch with the bases empty and 20 seconds to deliver a pitch when runners are on base. Furthermore, the batter has to be ready to hit with at least 8 seconds left on the clock. If the pitcher does not deliver the pitch within the allotted time, an automatic ball is recorded, and if the batter is not ready in the allotted time, an automatic strike is recorded (Castrovince, 2023).

Time Periods: Moneyball and Shift Era

The two periods that we will be examining need to be defined more in-depth. When we refer to the Moneyball Era, we are referring to the period in which teams started to look at advanced statistics and looked past the simple name value of players. This revolution was spurred by Billy Beane and the Oakland Athletics during the 2002 season and subsequently took over baseball. Following this revolution, poorer teams started to look at cheap players that could help them win by simply getting on base. The logic is simple: get on base more and win more games. However, does getting on base more lead to more wins during this period and is this reflected in player compensation?

What is the shift? According to Major League Baseball, “the shift is a term used to describe the situational defensive realignment of fielders away from their "traditional" starting points” (“Shifts: Glossary”). It is also important to note that infield and outfield shifts are defined

differently. The infield shift is defined by three infielders to either side of second base and an outfield shift is defined by three outfielders on either side of second base, which is quite rare. The shift became relatively popular in the early 2010s but exploded in the mid-late 2010s. According to MLB, "In 2017, lefty batters were shifted against on 22.1 percent of their plate appearances" ("Shifts: Glossary"). According to Sports Illustrated, the shift rate at the beginning of the 2022 season (about 37%) has "more than doubled in just 4 years" ("Shifts Increasing the Opener"). The graph depicted below from Baseball Info Solutions depicts the dramatic increase in the usage of the shift from 2010-2018. And as we know from the previous statistic, this number has increased even more since 2018 (Simon and Vigderman, 2018).

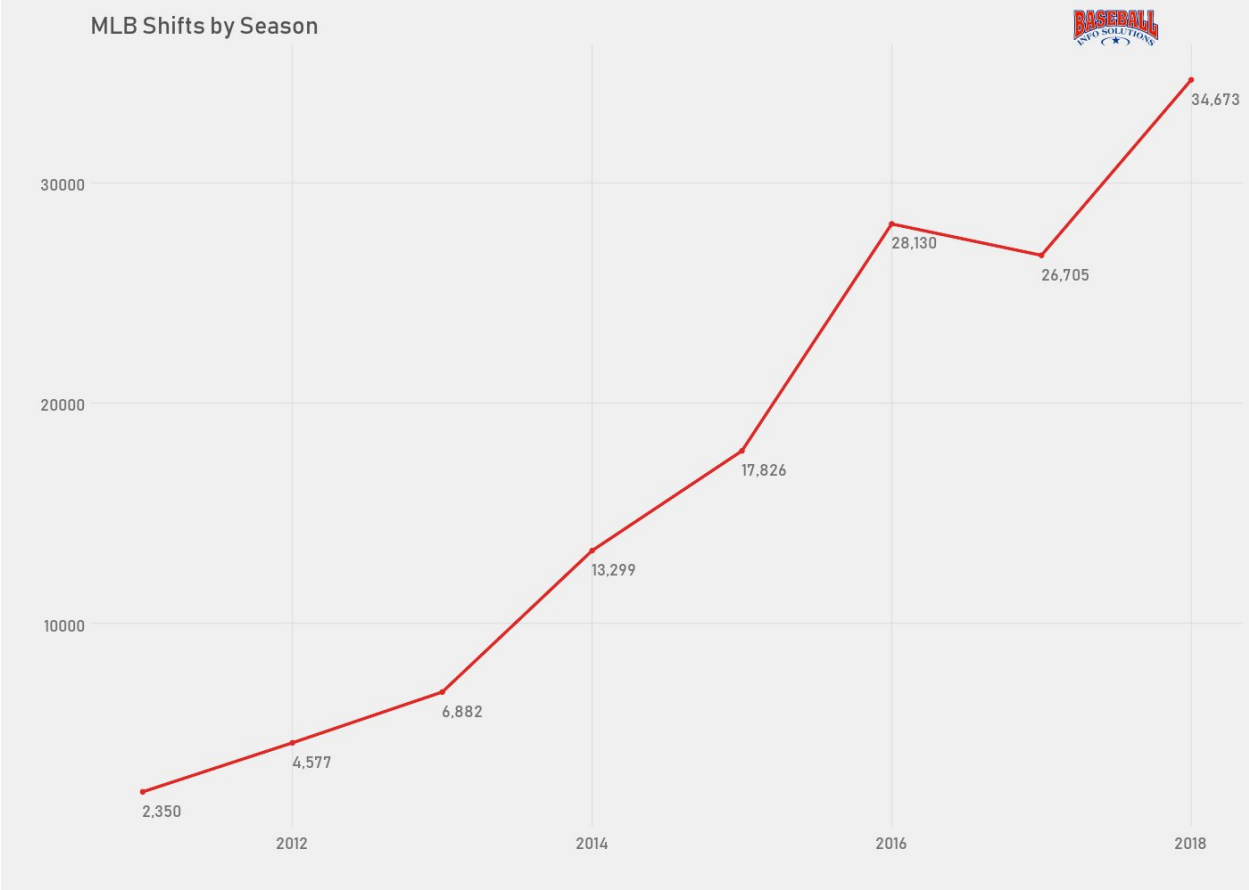


Figure 1: MLB Shifts By Season

Defining Statistics:

It is also important to further define the statistics that we will be using for our analysis. Some statistics are relatively basic while others are somewhat advanced. More recently, Major League Baseball has seen an uptick in the use of advanced analytics. These statistics help quantify certain aspects of the game that we could have never imagined to be possible. For example, a statistic known as wRC+ quantifies how many runs a player is creating while taking into account park or era effects. Statistics like this are changing the game and helping front offices acquire players that may fall under the radar. While it is important to look at these advanced statistics, we also need to pay attention to some of the more basic ones.

Basic Statistics:

The basic statistics that we will be using for our analysis include on-base percentage (OBP), slugging percentage (SLG), stolen bases (SB), home runs (HR), total plate appearances (TPA), and winning percentage (WP).

I. On-base percentage (OBP):

   A. OBP is the number of times that a player gets on base per plate appearance. The formula for OBP is as follows:

      1. $OBP = (Hits + BB + HBP)/(AB + BB + HBP + SF)$ where BB is walks, HBP is hit by pitch, AB is at-bats, and SF is sacrifice flies ("On-Base Percentage (OBP): Glossary"). Obviously, the more you get on base, the higher your OBP is.

II. Slugging Percentage (SLG):

   A. Slugging percentage is used to show the total number of bases a player tallies each at-bat ("Slugging Percentage (SLG): Glossary"). Different types of outcomes

have different weights. For example, a single holds less weight than a home run because a single is only one base while a home run is four. The equation is as follows:

1. $SLG = (1B + 2 * 2B + 3 * 3B + 4 * HR)/AB$ where 1B is a single, 2B is a double, 3B is a triple, HR is a home run, and AB is at-bats. A higher slugging percentage means you are tallying more bases in each at-bat, often signifying a large amount of extra-base hits (doubles, triples, home runs).

III. Stolen Bases:

A. Stolen bases are simply recorded when a player runs from one base to another successfully without being thrown out. A player can steal from first base to second base, second base to third base, and third base to home. However, players are not rewarded for stolen bases on errors, wild pitches, or passed balls ("Stolen Base (SB): Glossary").

IV. Home Runs:

A. According to Major League Baseball, a home run "occurs when a batter hits a fair ball and scores on the play without being put out or without the benefit of an error" ("Home Run (HR): Glossary"). Most often, a home run is hit fairly outside the park but sometimes, players can hit what is known as an inside-the-park home run. This occurs when the ball stays in the park but the player scores without any errors.

V. Total Plate Appearances (TPA):

A. A plate appearance occurs when a batter takes their turn at the plate. The difference between an at-bat and a plate appearance is an at-bat only occurs with certain results, but a plate appearance occurs every time there is an outcome recorded between the batter and the pitcher. Total plate appearances is the total number of plate appearances a player has in a season.

VI. Winning Percentage (WP):

A. Winning percentage is the number of games a team has won divided by the total number of games played. It is defined by this formula:

1. $WP = W/TG$ where W is wins and TG is total games

Advanced Statistics:

We are only using one advanced statistic for our analysis and that is defensive runs saved (DRS).

I. Defensive Runs Saved (DRS):

A. According to MLB, DRS "quantifies a player's entire defensive performance by attempting to measure how many runs a defender saved. It takes into account errors, range, outfield arm, and double-play ability" ("Defensive Runs Saved (DRS): Glossary"). We will not go in-depth on exactly how this is calculated but the more DRS a player has the better defender they are. However, it is important to note that there is no positional adjustment. So, more difficult positions are rated the same as easier positions.

Data:

To determine what matters to winning, we used data from the Lahman database ("Lahman's Baseball Database"). From here, we created variables like winning percentage (WP), on-base percentage (OBP), and slugging percentage (SLG) using the formulas described above. We analyzed two periods in baseball history, the Moneyball Era (2003-2010) and the Shift Era (2011-2022). We started in 2003 because there was no DRS metric available pre-2003. In this analysis, we also decided to add in a different time period in which the shift was heavily used, spanning from 2015-2022. So, our data here includes statistics from the years 2003-2022. While each period is a large time frame, we believe that this allows for any significant trends in baseball to become evident in the regression. Changes in baseball do not happen extremely quickly, so it is important to allow time for the full effects of the changes to take place. In the regression analysis to see what matters to winning, we included winning percentage (WP) as the dependent variable and added OBP, home runs (HR), stolen bases (SB), and defensive runs saved (DRS), and ran a fixed effects regression. We also added a variable that accounted for year-to-year differences.

In order to analyze player compensation, we used player offensive statistics data from Baseball Savant and player salary data from Cot's Baseball Contracts: A clearinghouse for MLB salary and payroll details ("Baseball Savant: Trending MLB Players, Statcast and Visualizations"; "Cot's Baseball Contracts"). The player offensive statistics were from 2000-2022 and we did not include any player with less than 150 plate appearances. The free-agent salary data is from the years 2001-2023 and it includes every free agent from each year. We then matched each free agent player with their stats from the previous season. For example, a 2023 free agent was matched with their stats from 2022. This gives us the simplest

way to value the free agent's contracts, by comparing their statistics from the previous season. For the analysis, we used the log of the AAV (Annual Average Value) of the free agent contract as the dependent variable, which is the log of the average dollar amount per year of the contract. We decided to use three distinct periods, which is different from our first analysis. Our three periods are the Moneyball Era (2000-2007), the Pre-Shift Era (2007-2015), and the Shift Era (2015-2022). Even though shifting occurred during the Pre-Shift Era, it was not as commonly used then as it was from 2015-2022. We see in Figure 1 that the number of shifts skyrocket in 2015, which is why we wanted to use this time frame for the Shift Era in this analysis. During the Pre-Shift Era, we expect to see some elements of the Moneyball Era still lingering and also some elements of the Shift Era developing. Player compensation often varies from season to season, so it is important to capture these changes with smaller and more distinct time frames.

Results:

We begin with a model of winning percentage. After running regressions for each time period (Moneyball Era, Shift Era, and our subgroup within the Shift Era from 2015-2022), we then used the same model but only for teams with a winning percentage above .500. This was done to see if there are any differences in what matters to winning for teams that actually win. We discuss our results after each model and their implications. Following this, we used a model of player compensation for each time period (Moneyball Era, Pre-Shift Era, and Shift Era) to see if what matters to winning is present in player compensation as well.

Winning Percentage Analysis:

Table 1: Summary Statistics of Variables in Model 1

Variable	Obs	Mean	Std. Dev.	Min	Max
WP	600	.4999487	.0747272	.2654321	.7166666
OBP	600	.3241718	.0142242	.2806987	.3655566
HR	600	168.8867	42.38921	51	307
SB	600	86.685	31.77576	14	200
DRS	600	2.958333	40.62853	-120	129

Depicted above is the summary statistics for the variables we will be using in the winning percentage model. One interesting note is that the mean for WP is very close to .500, meaning that each year there are a handful of teams above and below .500. A small standard deviation is present here, meaning that most teams do not deviate very far away from the mean. On the other hand, the home run (HR) variable has a large standard deviation of around 42. This shows that not all teams hit home runs at very high rates and this could definitely affect their winning percentage. DRS also has a large standard deviation, indicating that teams play different levels of defense (some really bad and some really good). DRS also has a mean of close to 3, showing that the average team over this time frame played "good" defense.

The final model used to see what matters to winning in each period is as follows:

$$WP_{it} = \beta0 + \beta1 * OBP_{it} + \beta2 * HR_{it} + \beta3 * SB_{it} + \beta4 * DRS_{it} + year_t + \varepsilon_{it}$$

Where WP = winning percentage, OBP = on-base percentage, HR = home run, SB = stolen bases, DRS = defensive runs saved, and ε is the error term. We include year indicators to account for year-specific effects.

Model 1: Regression Results for WP as Dependent Variable

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	Model 1.1	Model 1.2	Model 1.3	Model 1.4	Model 1.5	Model 1.6
On-Base Percentage	2.539***	2.829***	2.899***	0.870***	1.388***	1.446***
	(.299494)	(.3430769)	(.454414)	(.2782154)	(.3072055)	(.5027081)
Home Runs	0.000658***	0.000867***	0.000839***	0.000314***	0.000482***	0.000639***
	(.0001127)	(.0001448)	(.0001579)	(.00008)	(.0001304)	(.0001539)
Stolen Bases	0.000266*	-0.000160	-8.26e-05	0.000290**	-0.000136	0.000178
	(.0001354)	(.0001328)	(.0001698)	(.0001308)	(.0001359)	(.0002394)
Defensive Runs Saved	0.000478***	0.000661***	0.000570***	0.000162**	0.000291***	0.000221**
	(.0001014)	(.0000747)	(.0000897)	(.0000783)	(.0000929)	(.0001072)
Constant	-0.481***	-0.521***	-0.545***	0.184*	0.0409	-0.0448
	(.0979962)	(.1009324)	(.1292009)	(.0942497)	(.0947874)	(.1459198)
Observations	240	360	240	124	174	116
R-squared	0.396	0.583	0.519	0.222	0.334	0.357

Robust Standard Errors in parentheses
Year Fixed Effects are included in each regression
*** p<0.01, ** p<0.05, * p<0.1

Model 1.1: Moneyball Era (2003-2010)

OBP, HR, and DRS are all statistically significant, each at the 0.01 level. So is SB, but only at the 0.1 level. This model gives us an R-squared value of 0.396, indicating that our dependent variables explain about 39.6% of the variability in winning percentage for the Moneyball Era. OBP has the highest coefficient of 2.54, which means that for every one-unit

increase in OBP, a team's WP increases by 2.54 percentage points, ceteris paribus. This indicates the importance of OBP during the Moneyball Era.

Model 1.2: Shift Era (2011-2022)

The regression results for the Shift Era produce similar results. Here, the OBP coefficient, HR coefficient, and DRS coefficient all increase relative to the Moneyball Era. Also, our R-squared value increases to 0.583. However, the SB variable becomes statistically insignificant. During the Shift Era, we see a lot less emphasis on stolen bases and more emphasis on home runs and extra-base hits (hence the increase in the HR coefficient and the now statistically insignificant SB variable). It became much harder to get on base because of the intensive use of shifts. So, this also explains why the OBP coefficient increased because it became harder to get on base and when players did get on base, it became more valuable to winning.

Model 1.3: Subgroup of Shift Era (2015-2022)

Here, we really focused on when the shift became extremely popular in the MLB and we see some different results. Once again, the OBP, HR, and DRS are all statistically significant and SB is statistically insignificant. Also, the DRS coefficient decreases from Model 1.2. This makes intuitive sense because, around this time, teams really started to focus on power-hitting and less on defense. With the shift, defense was not as important because players did not have to use their range or arm strength as much, allowing teams to place more importance on hitting. The OBP coefficient once again increases and makes sense using the intuition provided beforehand. Also, here the OBP coefficient is the highest out of all the regressions at 2.899.

Model 1.4: Moneyball Era (2003-2010 teams above .500)

Now, we wanted to see what "winning" teams (teams with a WP > .500) thought was important to winning. While it is important to see what all teams value, special attention should

be paid to “winning” teams, as the goal in every sport is to win. So, do winning teams value different things? To find out, we ran the same regression but only analyzed these “winning” teams. The results here are much different.

To start, our R-squared coefficient decreases to about 0.222 or 22.2%. This suggests that a lot of other factors help these “winning” teams. Some are immeasurable, like a favorable division, an easy travel schedule for the whole year, or even the fans. Others are measurable, like the team’s payroll. Even though the R-squared is quite low here, we think that the results provide some helpful insight.

Most notably, SB becomes statistically significant at the 0.05 level, which was not the case for the Moneyball Era for all teams. This tells us that teams that winning teams seem to be stealing bases more successfully during the Moneyball Era. Also, the DRS coefficient decreases by a lot and is almost statistically insignificant at the 0.05 level.

Model 1.5: Shift Era (2011-2022 teams above .500)

The OBP coefficient increases a lot from the Moneyball Era, which again makes sense because the shift has made it much more valuable to reach base. The SB variable is statistically insignificant in this regression. This tells us that stolen bases do not matter much in the Shift Era for winning teams, which is what we expected.

Model 1.6: Shift Era (2015-2022 and teams above .500)

When analyzing “winning” teams, the OBP coefficient is the highest in this era, which is consistent with Models 1.1-1.3. Also, SB is once again statistically insignificant and DRS is statistically significant, but only at the 0.05 level. Interestingly, for “winning” teams, defense does not seem to be as important. It has always been thought defense wins championships. While

the coefficient is never statistically insignificant for “winning” teams in any era, it produces the lowest coefficients and larger p-values.

Player Compensation Analysis:

The previous section shows what is important in predicting a team’s winning percentage. We now turn to estimate a model of player compensation to test whether what is important to winning percentage is compensated accordingly in player salaries.

Table 2: Summary Statistics of Player and Salary Statistics

Variable	Obs	Mean	Standard Deviation	Minimum	Maximum
AAV	393	9012583	7062374	475000	40,000,000
Log AAV	393	15.681	0.881	13.071	17.504
Total plate appearances	415	587.306	91.525	192	750
Slugging Percentage	415	0.447	0.071	0.268	.863
On Base Percentage	415	0.342	0.034	0.266	.515
Total stolen bases	415	8.699	10.446	0	58

TPA: total plate appearances, SLG: Slugging Percentage, OBP: On-base Percentage, TSB: Total Stolen Bases

Above is a table of the summary statistics of the variables we used in our analysis of player compensation over three periods: Moneyball Era (2000-2007), Pre-Shift Era (2007-2015), and the Shift Era (2015-2023).

The final model used to see what matters to winning in each period is as follows:

$$LAAV_{it} = \beta 0 + \beta 1 * TPA_{it} + \beta 2 * SLG_{it} + \beta 3 * OBP_{it} + \beta 4 * TSB_{it} + Year_{t} + \varepsilon_{it}$$

Model 2: Regression Results for Log AAV as Dependent Variable

VARIABLES	(1) Moneyball Era	(2) Pre-Shift Era	(3) Shift Era
Total Plate Appearances	0.00418***	0.00315***	0.00257***
	(0.00107)	(0.000893)	(0.000861)
Slugging Percentage	5.064***	6.273***	3.496***
	(1.056)	(1.049)	(1.020)
On-Base Percentage	2.445	4.018*	9.531***
	(2.199)	(2.045)	(2.081)
Total Stolen Bases	0.00761	0.0103**	0.0152**
	(0.00573)	(0.00438)	(0.00671)
Constant	9.585***	9.394***	9.951***
	(0.733)	(0.649)	(0.671)
Observations	111	146	136
R-squared	0.573	0.562	0.537

Standard errors in parentheses
Year Fixed Effects are included in each regression
*** p<0.01, ** p<0.05, * p<0.1

In the Moneyball Era, only total plate appearances and slugging percentage are statistically significant. This confirms that on-base percentage was overlooked in terms of player compensation compared to slugging percentage, which is consistent with what was found in the original Moneyball study.

In the Pre-Shift Era, every variable becomes statistically significant. TPA and SLG are statistically significant at the 0.01 level, TSB is significant at the 0.05 level, and OBP is significant at the 0.1 level. The SLG coefficient increases from the first period, and the OBP coefficient almost doubles from the first period. The TSB coefficient also increases significantly from the Moneyball Era. All of this means that the Moneyball analysis was correct. On-base percentage was overlooked in player compensation in the early 2000s and corrected itself. This is shown by OBP being statistically significant in the Pre-Shift Era. Also, TSB became statistically significant, which means that teams might be looking into what makes them win and compensating players accordingly. The coefficient for SLG is larger than the coefficient for OBP meaning during the Pre-Shift Era SLG was compensated for more than OBP in players' salaries.

Something interesting happens in the Shift Era. All of the variables are statistically significant. SLG, OBP, and TPA are significant at the 0.01 level and TSB is still significant at the 0.05 level. So, OBP became as important to salary as SLG. Also, the coefficients for SLG and OBP flipped. The coefficient for OBP is now significantly greater than the coefficient for SLG, which is the opposite of the previous two periods. This could mean OBP is now more important to salary than SLG in the Shift Era because it was much more difficult to get on base. Also, the coefficient continued to increase for TSB, meaning total stolen bases are becoming more important to player salary than the previous two periods.

Discussion:

After analyzing the two different periods in MLB's history to see what matters to winning, one conclusion is evident: on-base percentage reigns supreme. The OBP coefficient in all the regressions was significant and had a large impact on winning percentage (WP). While one may expect the OBP coefficient to be the highest in the Moneyball Era because of the focus on OBP, this is not what we see. The OBP coefficient is highest in Model 1.3 (the prime Shift Era for all teams) with a coefficient of almost 3. This means a 1 percentage point increase in OBP leads to a 3 percentage point increase in winning percentage, ceteris paribus. Why might this be? The shift made it much harder for players to get on base by hitting the ball in play. This raises the importance of walks and hit-by-pitches, which increases one's OBP.

Another trend noticed in our analysis is the insignificance of stolen bases (SB). In fact, SB was only significant once in all of our models at the 0.05 level and that is Model 1.4 (Moneyball Era & WP > .500) and is only significant twice at any reasonable level (both in Moneyball Eras). This makes a lot of sense given the context of our analysis. We also see that

home runs have higher coefficients compared to other periods when the SB variable is insignificant. This indicates a tradeoff that occurred in the MLB during these time periods. During the Moneyball Era, teams were more concerned with getting on base and then when getting on base, stealing bases. This seems to have changed during the Shift Era. Since it was harder to get on base, teams tried to find players that could help avoid the effects of the shift. One way around the shift is the home run. However, when a player hits a home run, they do not have the opportunity to steal a base. Hitting more home runs likely decreases the amount and value of stolen bases, which is what we see in our models. We also see the coefficient for home runs increase in every period for teams above .500, indicating the increased significance of power hitters over time.

Another important factor in a team's winning percentage is defensive runs saved (DRS). This is the only defensive metric that we focused on for these models, so it is important to analyze. DRS was statistically significant for all of our models at the 0.05 level and below. However, defense seems to be less important for "winning" teams in the Shift Era and the Prime Shift Era. This introduced another trend in the MLB. Once teams started to heavily shift, the value of defense decreased. When shifting, players are not required to be as athletic and mobile as before because they are in such close proximity to other fielders. So, these teams can afford to put a player in the field that is not a great fielder but is a good hitter, specifically a power hitter. And based on our results, "winning" teams employed this strategy, as seen in the coefficient decrease for all teams and "winning" teams during these eras. This also plays into the importance of home runs as discussed before. Now that players do not have to be as good defensively, teams can acquire more power-driven players.

The salary side of the analysis similarly follows the results found in the winning percentage analysis. OBP becomes valued higher in salaries as time goes on, just as it gets more important to winning. SLG is statistically significant in all periods just like home runs are all statistically significant in the winning percentage models. One thing that is interesting is how stolen bases are statistically significant in the final period for salaries but not for winning percentages. This is possibly because stolen bases may not be as important for winning anymore, but players will be compensated accordingly if they can steal bases. And with the new rules, maybe stolen bases will become more important to winning again. This could be an area of future study. Maybe it is speed that is compensated for in salary, and faster players can obviously steal more bases, which is why stolen bases would be significant in the player compensation model but not the winning percentage model.

Conclusion:

What do our results mean for the 2023 MLB season and beyond? There are many implications for our analysis and the future. OBP became more and more significant as the shift was implemented into the game. We believe that this trend will continue and OBP will become increasingly important in a team's winning percentage and a player's salary. We also expect to see stolen bases become significant. While stolen bases were insignificant for most models in the winning percentage models, it was significant for most of the models in player compensation. While this is contradictory, it makes sense because it became more valuable for players to steal bases because it happened less with the shift. So, players were paid accordingly. With the bigger bases, pitch clock, pickoff limit, and the shift ban, we expect to see stolen base numbers increase dramatically for the foreseeable future as well as their contributions to winning and player

compensation. We also expect to see defense play more of a role for teams. Since the shift is now banned, this requires more athleticism for players and therefore, for them to be better defenders. The 2023 season in the MLB will be an extremely interesting one. We could maybe see the style of play completely flip on its head. We have already seen some of these changes come to fruition during the 2023 free agency market. Trea Turner, a speedy shortstop, signed with the Philadelphia Phillies for $300 million. He consistently ranks at the top for stolen bases each year. We also see Carlos Santana signed with the Pittsburgh Pirates. He was one of the most shifted players in baseball and has not had good offensive production in the last few years. The Pirates might think that because of the shift ban, he will have a good year. A few years from now, an analysis of this same style could help answer the question of whether the new rules have truly changed the game.

While our models do tell an interesting story about the MLB and its history, it is also important to acknowledge some limitations of this study. One is that we completely ignored the pitching side of baseball in Model 1. Pitching is another extremely important aspect of baseball and it is well known that good pitching leads to more wins. Focusing on hitting and defense for what matters to winning does help explain some of the variation in winning percentage but not all of it. This could explain some of the fairly low R-squared values for some of the models. Also, we do not account for defense in Model 2 and defense definitely does play a factor in player compensation. Nevertheless, we believe that our analysis helps to provide some insight into how the game of baseball has changed both from a team perspective and a player compensation perspective and how the new rule changes could potentially change the game once again.

Bibliography:

"Baseball Savant: Trending MLB Players, Statcast and Visualizations." *Baseballsavant.com*, https://baseballsavant.mlb.com/#.

Castrovince, Anthony. "Pitch Timer, Shift Restrictions among Announced Rule Changes for '23." *MLB.Com*, 13 Feb. 2023, www.mlb.com/news/mlb-2023-rule-changes-pitch-timer-larger-bases-shifts.

Congdon-Hohman, J. M., & Lanning, J. A. "Beyond Moneyball: Changing Compensation in MLB." *Journal of Sports Economics*, vol 19, no. 7, 2017, pp. 1046–1061.

"Cot's Baseball Contracts." *Cots Baseball Contracts*, legacy.baseballprospectus.com/compensation/cots/.

"Defensive Runs Saved (DRS): Glossary." *MLB.com*, https://www.mlb.com/glossary/advanced-stats/defensive-runs-saved.

Demmink, Herman. "Value of Stealing Bases in Major League Baseball: 'Stealing' Runs and Wins." *Public Choice*, vol. 142, no. 3/4, 2010, pp. 497–505. *JSTOR*.

Hakes, Jahn K., and Raymond D. Sauer. "An Economic Evaluation of the Moneyball Hypothesis." *Journal of Economic Perspectives*, vol. 20, no. 3, Summer 2006, pp. 173-185.

"Home Run (HR): Glossary." *MLB.com*, https://www.mlb.com/glossary/standard-stats/home-run#.

"Lahman's Baseball Database." *SeanLahman.com*, 20 Feb. 2023, https://www.seanlahman.com/baseball-archive/statistics/.

"On-Base Percentage (OBP): Glossary." *MLB.com*, https://www.mlb.com/glossary/standard-stats/on-base-percentage#.

"Shifts: Glossary." *MLB.com*, https://www.mlb.com/glossary/statcast/shifts.

Shorey, Christopher B., "Defensive Ability and Salary Determination in Major League Baseball" (2016). *Honors Theses*. Paper 940.

Simon, Mark, and Alex Vigderman. "2019 Bill James Baseball Handbook: 'Shift Update.'" *Sports Info Solutions*, 1 Nov. 2018, https://www.sportsinfosolutions.com/2018/11/01/2019-bill-james-baseball-handbook-shift-update/.

"Slugging Percentage (SLG): Glossary." *MLB.com*, https://www.mlb.com/glossary/standard-stats/slugging-percentage#.

Sports Illustrated. "Shifts Increasing the Opener." *SI*, 26 Apr. 2022, https://www.si.com/mlb/2022/04/26/shifts-increasing-the-opener.

"Stolen Base (SB): Glossary." *MLB.com*, https://www.mlb.com/glossary/standard-stats/stolen-base.

"Total Plate Appearances (TPA): Glossary." *MLB.com,* https://www.mlb.com/glossary/standard-stats/plate-appearance#